

EDUCATION

Carnegie Mellon University
B.S. in Computer Science, Minor in Mathematical Sciences

2014 - 2018

EXPERIENCE

Qatar Computing Research Institute
Software Engineer

August 2018 - Present
Doha, Qatar

- Worked on the full-stack development of Rayyan (rayyan.qcri.org), a popular online research platform for conducting systematic reviews. Rayyan currently has 80K+ registered users and 30K+ monthly active users.
- Implemented a custom PostgreSQL-backed storage layer as a replacement to the existing Solr-based search cluster, eliminating indexing times and improving application response times by 50%.
- Designed and developed a load balancer for the above-mentioned storage layer to distribute traffic and data evenly among horizontally-scaled replicas.
- Responsible for containerizing and deploying various Rayyan microservices using Docker and Kubernetes.

Carnegie Mellon University Qatar
Research Consultant

May 2018 - Present
Doha, Qatar

- Hired as a consultant by Prof. Ihab Younis to assist with the computational demands of his research lab.
- Built a genomics pipeline that combines various bioinformatics tools for analyzing RNA-Seq data at scale. Includes a custom component for computing splicing efficiency and read coverage across specific genomic regions.
- Built a dashboard to visualize the results of RNA-Seq analysis on breast cancer samples, as part of a deliverable for Prof Younis's NPRP research grant.
- Worked on cluster analysis of the TCGA-BRCA dataset using expression levels of U12-intron containing genes.

PROJECTS

rayyan-store: A PostgreSQL-backed data store for [Rayyan](https://rayyan.qcri.org) which abstracts data storage and allows for both read and write heavy queries to application data.

rayyan-store-proxy: A Go-based load balancer that serves traffic to horizontally scaled rayyan-store replicas.

splicing-analysis: Computes read coverage across exon-intron, intron-exon and flanking exon-exon junctions. Additionally, computes a psi value used to extrapolate splicing efficiency of introns. Written in Python.

rnaseq-pipeline: A Nextflow pipeline for aligning, mapping & estimating abundances of RNA-Seq data.

GenomeDB: A React-based web portal for visualizing RNA-Seq data generated from YounisLab's breast cancer research. Accessible at genomedb-ihylab.herokuapp.com.

TECHNICAL SKILLS

Computer Languages

Python, Ruby, TypeScript, Go, C, C++

Backend

Ruby on Rails, Node.js, Flask, PostgreSQL, MongoDB

Frontend

jQuery, React

Devops

Docker, Kubernetes, Azure, Heroku